

Quantifying Model, Label, and Methodological Contributions to Variance in Real-Time Sepsis Onset Prediction

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I. Introduction: Discrimination Does Not Guarantee Clinical Utility

- Sepsis is associated with about **11 million deaths a year**, nearly one in five deaths worldwide [1]; in the ICU every hour of delayed treatment raises the risk of death [2].
- Wang et al. [3] reviewed **91** real-time sepsis prediction models:

Metric	Internal	External
Median AUROC	0.811	0.783
Median Utility Score	+0.381	−0.164

- A negative Utility Score means performance worse than issuing no alerts at all. AUROC did not detect this decline, because it measures ranking, not whether alerts are timely or useful.
- The same gap appears in deployment: Epic Sepsis Model v2 across 227,091 encounters, AUROC 0.82–0.92 with PPV 0.13–0.26 [4].

Comparing models alone does not show whether reported performance is robust or clinically useful.

II. Background: Limitations of the Conventional Approach

Choose one Sepsis-3 definition → use one train/test split → report AUROC → compare models.

1. **The sepsis label is not fixed.** Different ways of applying Sepsis-3 to the same database gave 867–2,178 cases [5].
2. **The data split can inflate results.** Sepsis prediction suffers the largest drop of any clinical task under a temporal shift [6].
3. **AUROC may not reveal poor clinical utility.** A model can keep a high AUROC while its Utility Score falls below the no-alert strategy [3].
4. **The label partly reflects clinical practice.** Models may learn patterns of sepsis recognition and treatment rather than physiological deterioration alone [7].

This thesis

This thesis evaluates model class, label definition, data split and evaluation metric as separate **sources of variation**. Each contribution is estimated with a **bootstrap confidence interval**, so their relative sizes can be compared directly.

III. Research Question and Pre-Registered Hypotheses

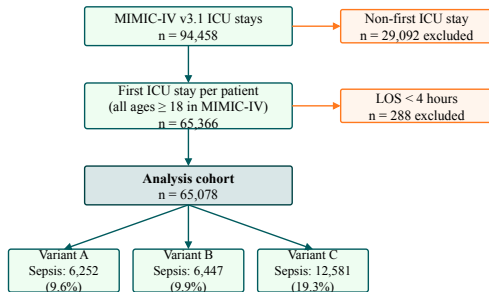
Research Question

In real-time sepsis prediction on MIMIC-IV [8], how much of the reported performance variation comes from (a) the choice of model, and (b) the researcher's decisions: how sepsis is defined, how data is split, when prediction is made, and how performance is measured?

- **H1 (Label dominance):** changing the sepsis definition affects AUROC at least as much as changing the model.
- **H2 (Coefficient instability):** at least one feature changes direction, or magnitude $\geq 2\times$, across label variants.
- **H3 (Treatment leakage):** restricting evaluation to before treatment reduces AUROC by ≥ 0.03 .
- **H4 (Split optimism):** temporal testing gives ≥ 0.02 lower AUROC than random testing.
- **H5 (Metric divergence):** Utility declines more than AUROC from internal to external validation.
- **H6 (Model-class null):** the primary statistical model is not outperformed by gradient-boosted trees by more than 0.02 AUROC.

The six hypotheses form one Holm-corrected family ($m = 6$). H2 and H3 keep their places in the family but produce no p-value; the reported H5 contrast is a post-hoc substitute for the pre-registered estimand. **Scope:** the models estimate the hazard of onset in the *current* hour, i.e. detection at the recorded onset, not advance warning.

IV. Materials: Data and Cohort



- **Development:** MIMIC-IV v3.1 [8], a single centre (BIDMC, Boston).
- **Inclusion:** age ≥ 18 , first ICU stay per patient, length of stay ≥ 4 h $\Rightarrow$ 65,078 **stays**.
- **External:** eICU-CRD v2.0 [9], over 200 US ICUs, 181,589 unit stays; models applied **frozen, without retraining**.
- **Person-hour panel:** 72 h horizon; discharge and death are competing events, not censoring.
- **22 covariates** (6 vitals, 5 labs, 2 vasopressor flags, 3 six-hour slopes, 6 test-ordered indicators), 29 terms per outcome.

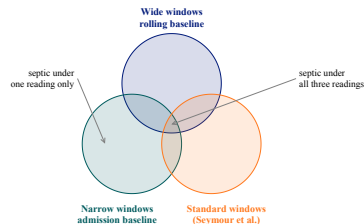
IV. Materials: Three Pre-Registered Sepsis-3 Label Variants

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline	Septic stays
A	Narrow	24 h	24 h	24 h	12 h	Admission	6,252 (9.6%)
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission	6,447 (9.9%)
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h	12,581 (19.3%)

ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which a SOFA rise of ≥ 2 must be observed.

- All three: **suspected infection** (antibiotic paired with a culture inside the windows) **and** a SOFA rise of ≥ 2 points [5], [10].
- **Variant B is the primary analysis**; A and C are pre-specified alternatives.
- Onset is assigned at the suspicion time t_{susp} . The SOFA criterion decides which stays are septic, not when onset occurs.

A and B have near-identical prevalence but agree on which stays are septic at only $\kappa = 0.267$.



V. Methods: Models and Validation Architecture

Primary model: dynamic discrete-time competing-risks hazard

Multinomial pooled logistic regression over person-hours (sepsis / discharge / death), natural-spline baseline in elapsed hours [11], [12]. The unpenalised fit is completely separated, so a small ridge penalty is required for identification.

Comparators: gradient-boosted trees (XGBoost), Cox proportional hazards, NEWS2, qSOFA and SIRS [13]. XGBoost uses fixed hyperparameters; the number of boosting rounds is chosen by early stopping on a 15% stay-level holdout from the *training* data.

Split	How	Why
Internal (primary)	Time-based: train 2008–2016, test 2017–2019 [6]	Primary internal evaluation
Internal (secondary)	Random 75/25 patient-level split (seed 42)	Split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0, no retraining [9]	Generalisability (H5)

All splits are patient-level: no patient appears in both training and test sets.

V. Methods: Metrics and Statistical Inference

Primary metric: adapted PhysioNet-style Utility Score [14]

$$\tilde{U} = \frac{U - U_{\text{inact}}}{U_{\text{opt}} - U_{\text{inact}}} \quad \tilde{U} = 1 \text{ perfect, } \tilde{U} = 0 \text{ no-alert strategy, } \tilde{U} < 0 \text{ worse than no alerts}$$

- Rewards early true alerts in $[t_{\text{onset}} - 6, t_{\text{onset}}]$, penalises late and false ones.
- Reported at the 0.5 cut-off **and** at its maximum over a 60-point alert-rate grid. Alert burden is false alerts per true alert, compared with the published benchmark of 1.4 [15].
- AUROC, AUPRC and calibration are reported alongside.

Inference and reporting

- Stay-level cluster bootstrap, $B = 2,000$; H1–H6 Holm-corrected ($m = 6$, FWER 0.05); exploratory families Benjamini–Hochberg.
- Pre-registered hypothesis set; every post-hoc protocol amendment is identified explicitly; TRIPOD+AI and PROBAST+AI reporting [16], [17].

VI. Results: Internal Temporal Performance (Variant B)

Model	AUROC	95% CI	Utility (best)	Alert burden at best
Primary	0.7590	[0.7434, 0.7749]	0.025	47.3
GBT	0.7424	[0.7265, 0.7583]	0.031	51.8
Cox	0.6444	[0.6271, 0.6614]	0.000	∞
NEWS2	0.6415	[0.6248, 0.6589]	0.000	∞
SIRS	0.6114	[0.5950, 0.6281]	-0.035	135.8
qSOFA	0.6091	[0.5934, 0.6254]	-0.252	248.8

Temporal test set 2017–2019: 548,827 person-hours, 1,040 onsets. Utility (best) = maximum over the threshold grid; ∞ = the best strategy is to raise no alert. Benchmark burden: 1.4 [15].

- The marginal confidence intervals of the primary model and GBT overlap, but the paired comparison favours the primary model by 0.0165 AUROC (H6). Both have higher AUROC than the four other comparators.
- The primary model is well calibrated under Variant B: calibration slope 1.055 (ideal = 1).

VI. Results: The Variance Decomposition

Factor varied (others fixed)	Δ AUROC	95% CI	H
Model class (label B, temporal)	0.1499	[0.134, 0.171]	n/a
External transport (primary, label B)	0.0630	[−0.0215, 0.1563]	H5
Label definition (primary, temporal)	0.0123	[0.003, 0.032]	H1
Split method (primary, label B)	−0.0165	[−0.0372, 0.0040]	H4
GBT — primary (label B, temporal)	−0.0165	[−0.0308, −0.0012]	H6

Ranges, not variance components: they cannot be summed or expressed as shares of a total. The model-class range spans six heterogeneous specifications, the label range three Sepsis-3 readings. Despite this asymmetry, the GBT–Cox difference (0.7424 vs 0.6444) is about eight times the whole label range.

- **Model class** produces the largest AUROC range of the factors examined.
- External transport has the second-largest point estimate, but its wide interval includes zero, so its relative position is uncertain.
- In this analysis, **label definition and split method produce the smallest changes in AUROC.**

VI. Results: Corrected Confirmatory Verdicts

H	Contrast	Est.	95% CI	p_{Holm}	Verdict
H1	Label spread — model spread	−0.1375	[−0.1623, −0.1083]	1.000	NOT confirmed
H2	Unstable covariates (count)	6	n/a	n/a	Descriptive
H3	Pre-treatment onsets (count)	0	n/a	n/a	Definitional
H4	Random — temporal AUROC	−0.0165	[−0.0372, 0.0040]	1.000	NOT confirmed
H5 [†]	Internal — external AUROC	0.0630	[−0.0215, 0.1563]	0.551	Not evaluable as pre-registered; substituted AUROC contrast inconclusive
H6	GBT — primary AUROC	−0.0165	[−0.0308, −0.0012]	1.000	H6 not rejected

[†] Post-hoc substituted contrast occupying H5's slot: the proportional utility-versus-AUROC contrast cannot be formed, because there is no internal Utility Score to decline from.

- **H1 not confirmed:** model spread 0.1499 vs label spread 0.0123; the estimate lies in the opposite direction and its interval excludes zero.
- **H4 not confirmed:** opposite direction; the random split scores *lower* (0.7424 vs 0.7590) for every model.
- **H6 not rejected:** primary 0.7590 vs GBT 0.7424. The upper limit lies inside the 0.02 margin, so one-sided **non-inferiority**, not two-sided equivalence.

VI. Results: H3, the Empty Pre-Treatment Window

Status: (NOT EVALUABLE AS PRE-REGISTERED), because its contrast cannot be formed.

Proposition (proved in the thesis)

In Variants A, B, C and E, onset is assigned to the *later* event of the antibiotic–culture pair, so onset can never precede the treatment anchor. The pre-treatment window therefore contains no onsets **by construction**, not by observation.

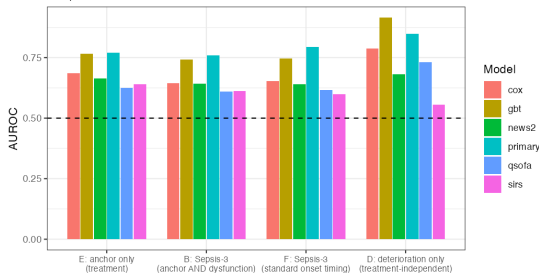
Var	Label	Pre-treatment person-h	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	undefined
F	Sepsis-3 Early-Onset	178,303	71	0.8081 / 0.8018
D	Deterioration	147,299	1,362	0.8498 / 0.9360

- **Variant F** (post-hoc): B's membership exactly, onset at the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$. 1,738 of 6,447 onsets move earlier (27.0%, median -14.0 h).
- Kamran et al.'s concern [7] is supported for case membership but needs qualification for onset timing. These AUROCs measure discrimination at the assigned onset hour; they do not demonstrate advance prediction.

VI. Results: Splitting the Sepsis-3 Conjunction

Labels E, D and F are declared protocol amendments: (EXPLORATORY, POST-HOC)

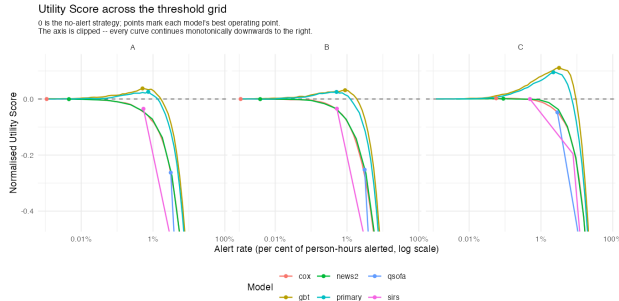
Splitting the Sepsis-3 conjunction into its two limbs
Temporal test set, unrestricted window; dashed line = chance



- **E (Anchor-Only)**, no SOFA criterion, matches B's onset hour in 100.0% of B's stays: SOFA does not change the onset hour.
- Anchor-attributable fraction:
 $\phi_{\text{anchor}} = 1.041 [0.990, 1.097]$ (primary, includes 1); $1.099 [1.037, 1.164]$ (GBT).
- **D (Deterioration)**, no treatment timestamp:
 $\kappa = 0.057$ with B. D's definition overlaps the model inputs, so its AUROC is not independent evidence of sepsis prediction.
- **F (standard onset timing)**: primary 0.7951 vs 0.7590; **not reproduced by GBT** ($\Delta = 0.0033$, $q = 0.559$).

For the primary model, changing the onset time matters more than removing the SOFA criterion. For GBT the timing effect is not statistically detectable.

VI. Results: Clinical Utility Across Alert Thresholds



- At the 0.5 cut-off the primary model scores 0.000: it never alerts there, a threshold artefact.
- Across the threshold grid, the highest Utility Score under Variant B is 0.031 (0.111 under any variant), at 51.8 **false alerts per true alert** against the published benchmark of 1.4 [15].
- Thresholds were chosen on the test set, so these values are optimistic and are not estimates of deployed performance.

VI. Results: External Validation on eICU-CRD

Var	Internal (primary)	eICU primary	eICU GBT	eICU NEWS2	Events
A	0.7713	0.6913	0.7106	0.5866	27
B	0.7590	0.6960	0.6815	0.5259	29
C	0.7695	0.7370	0.6877	0.5074	73

Sepsis-3 arm on the microbiology-covered sub-cohort: frozen models, no retraining.

- **The main limitation is incomplete label observability:** culture records restrict the Sepsis-3 analysis to 1.55% of eICU stays, leaving 29 Variant B events, below the 100-event floor.
- H5 substitute: degradation 0.0630 $[-0.0215, 0.1563]$, directionally consistent but inconclusive. Its causes are not decomposed.
- The antibiotic-only (6,140 events) and Variant D (52,535 events) sensitivity analyses use different targets and are reported separately. Both show lower external than internal discrimination.

VI. Results: Coefficient Instability and Equity

H2: coefficient instability (descriptive)

- 6 of 22 covariates unstable across labels: 2 by sign, 5 by magnitude.
- **Only 3 remain unstable after accounting for estimation uncertainty, all test-ordered indicators** [18]; the raw count is a loose upper bound.
- **Ablation** (post-hoc): dropping the 8 action-derived covariates costs -0.0076 AUROC (primary, B), while instability falls to 1 of 14. The AUROC reduction is small but distinguishable from zero.

Equity by label (exploratory)

- Subgroup AUROC: 6 contrasts clear the 100-event floor and **none survives** BH correction.
- Only two race categories meet the 100-event floor, one of which is *Unknown*, so no powered comparison between two recorded categories is possible.
- Label sensitivity *does* survive: Unknown $+3.01$ pp ($q = 0.006$), Unable to obtain $+3.49$ pp ($q = 0.009$).
- Both indicate missing demographic information, so the result is read as an association with **documentation completeness**, not as a group difference [19].

VII. Discussion: The Central Lesson and Limitations

The Sepsis-3 label on MIMIC-IV:

1. **Depends on clinical action for membership.** The empty pre-treatment window results from the pre-registered onset rule: standard timing moves 1,738 onsets earlier, 71 of them into the test set's pre-treatment window (onset-hour separability, not advance warning).
2. **Is sensitive to how it is defined.** Cases nearly double across definitions ($\kappa = 0.267$), yet AUROC moves by only 0.0123.
3. **Produces poor clinical utility at the evaluated thresholds.** No operating point comes near the 1.4 benchmark; the best under B costs 51.8 false alerts per true alert.

These limitations are not resolved by a more complex model: improving discrimination alone does not address limitations created by case definition and onset timing.

Key limitations

- **Current-hour estimand:** detection at the recorded onset, not advance warning.
- **External validation is limited by incomplete transport of the Sepsis-3 label:** 1.55% culture coverage, 29 events under B.
- **Single-centre development**, with a selected base cohort (first stays, ≥ 4 h).
- **The pre-registered onset rule is not the standard one;** F restores it post hoc.
- **Variant D is not a validated construct;** its components overlap the model inputs.
- **Post-hoc protocol amendments** are disclosed as such; the literature search is narrative, not systematic.

VIII. Conclusion: Contributions and Future Work

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- C1 Label and model-class spreads on the same test rows, with bootstrap confidence intervals: label definition substantially changes case membership ($\kappa = 0.267$) but changes discrimination only slightly (0.0123).
 - C2 Proof that the conjunction onset rule empties the pre-treatment window; standard timing places 71 test-set onsets there.
 - C3 Threshold-swept Utility Score: at most 0.031 under B, at 51.8 false alerts per true alert.
 - C4 6 of 22 covariates unstable across labels; 3 remain unstable after accounting for estimation uncertainty, all test-ordered indicators.
 - C5 Equity by label: differences are associated with missing demographic documentation; no powered comparison between recorded race categories.
 - C6 Pre-registered hypothesis set; seed-fixed, reproducible pipeline with disclosed post-hoc amendments; TRIPOD+AI reporting; reported values generated directly from the analysis outputs.
 - C7 Definitional vs. empirical label variance separated ($\kappa_{D,B} = 0.057$; $\phi_{\text{anchor}} = 1.041$).
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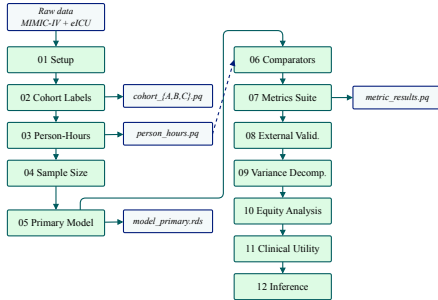
Future work: a sepsis definition grounded in objective physiology and adjudicated independently of the model's features; SEP-1 and CDC Adult Sepsis Event; prospective, protocol-blinded and multi-centre validation; utility-optimised training.

Label definition mainly changes which stays are septic and when onset is assigned; its effect on AUROC is small (0.0123).

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Appendix: The Twelve-Stage Pipeline



- One orchestration script runs stages 01–12; a single configuration module defines the file paths, identifiers and analytical thresholds.
- Reported results are generated directly from the pipeline outputs, reducing the risk of transcription errors between analysis and thesis.
- Frozen-run tag: frozen-2026-08-31.
- Code and reproducibility:
github.com/ahadarif007/sepsis-variance-decomposition

Thank you

Questions and discussion

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Pre-registered hypotheses · Post-hoc amendments disclosed

Code: github.com/ahadarif007/sepsis-variance-decomposition